SCAAA News Letter

Vol. 12 No. 18, Jan- Feb -2024

SAMANTA CHANDRA SEKHAR AMATEUR ASTRONOMERS' ASSOCIATION, BHUBANESWAR

Estt. 10.7.93, Regd. No. KRD/5907-125 of 1994-95

Asymptotic Giant Branch (AGB)

Bibhudutta Panda, Member, SCAAA

The asymptotic giant branch (AGB) is a region of the Hertzsprung–Russell diagram populated by evolved cool luminous stars. This is a period of stellar evolution undertaken by all low- to intermediate-mass stars (about 0.5 to 8 solar masses) late in their lives.

Observationally, an asymptotic-giant-branch star will appear as a bright red giant with a luminosity ranging up to thousands of times greater than the Sun. Its interior structure is characterized by a central and largely inert core of carbon and oxygen, a shell where helium is undergoing fusion to form carbon (known as helium burning), another shell where hydrogen is undergoing fusion forming helium (known as hydrogen burning), and a very large envelope of material of composition similar to main-sequence stars (except in the case of carbon stars). When a star exhausts the supply of hydrogen by nuclear fusion processes in its core, the core contracts and its temperature increases, causing the outer layers of the star to expand and cool. The star becomes a red giant, following a track towards the upper-right hand corner of the HR diagram. Eventually, once the temperature in the core has reached approximately 3×108 K, helium burning (fusion of helium nuclei) begins. The onset of helium burning in the core halts the star's cooling and increase in luminosity, and the star instead moves down and leftwards in the HR diagram. This is the horizontal branch (for population II stars) or a blue loop for stars more massive than about 2.3 $M_\odot$.

After completion of helium burning in core, the star again moves to the right & upwards on the diagram, cooling and expanding as its luminosity increases. Its path is almost aligned with its previous red-giant track, hence the name asymptotic giant branch, although the star will become more luminous on the AGB than it did at the tip of the red-giant branch. This stage of stellar evolution are known as AGB stars.

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Detection of Farthest Galaxies & its Impact on Understanding of the Universe

Aditya Sahasranshu, Member,SCAAA

*"With increasing distance, our knowledge fades & fades rapidly. Eventually, we reach dim boundary—the utmost limits of our telescopes. There, we measure shadows & we search among ghostly errors of measurement for landmarks that are scarcely more substantial. Search will continue". - **Edwin P. Hubble***

Our universe has gone through many phase transitions and one of them is where we saw the formation of the first luminous sources. Formed from the primordial gases prevalent in the universe and beginning in the Epoch of Reionization where the neutral hydrogen gases were ionized by these stars. To understand this better and understand the beginning of our universe we have been looking further and further into the past. In one of those endeavors using the James Webb Space Telescope (JWST), we discovered some of the most distant galaxies ever detected by humans.

With these discoveries, we are looking at our universe when it was just 330 million years young, or just 2% of its present age. The galaxies in question are the JADES-GS-z13-0, UNCOVER-z13, and UNCOVER-z12, GLASS-z12 which are some of the most distant galaxies ever observed, with redshift (z) of 13.2, 13.079, 12.393, and 12.117 respectively. Until now the most distant galaxies observed had $z \sim 11$, and substantial advancement in the detection and study of galaxies with $6<z<10$, gave information about the early star formation activities and the reionization process. The discovery of these galaxies has enabled us to look farther into the past of the universe and will also help us look for answers to many unanswered questions of extragalactic astronomy like the coalescence of the first cosmic structures, the formation of supermassive black holes, etc. Detection of UNCOVER-z13 and UNCOVER-z12 having mass $\sim 10^8$ $M_\odot$, was made possible because of the immense gravitational lensing caused by the ABELL 2477 galaxy cluster nicknamed Pandoras' Cluster, this galaxy cluster warped the space-time fabric, making it possible for these galaxies to be detected by the JWST.

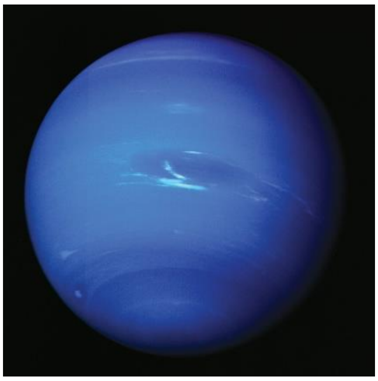
Just looking into the UNCOVER-z12 galaxy that was detected using the NIRSpec instrument of the JWST, it has a very interesting morphology and shows a disk-like shape, but there appear to be two clumps in the image captured, showing an extended size, this distributed morphology can be due to clumpy star-forming region, or an ongoing merger can also explain this morphology. As a clarification, these galaxies are observed as they were 13.4 billion years ago, so at present it would be very different and most likely a part of a supercluster. The two clumps are designated as two sources with Source-1 accounting for 2/3 of the total flux of the system. As said previously this could indicate a merger scenario which shows a hierarchical structure formation process where small structures collapse together to form a larger one. So, these detections are consistent with the previous theories about early-type galaxies having low mass $\sim 10^8$ $M_\odot$, young star formation, rapid assembly, low metallicity, etc.

As said in the beginning to understand our universe, we will have to keep looking into the past until we detect some of the first lights of the very first stars, and this detection and study of the farthest galaxies and possible future detection of other distant galaxies will give us much-needed insight, that'll help us answer many unanswered mysteries and further question our understanding of the universe.

The Great Dark Spot on the planet Neptune

Golak Bihari Pradhan, Jt. Sec., SCAAA

It is known that the largest planet of the solar system, the Jupiter has a Great Red Spot. The high pressure region in the southern hemisphere atmosphere of the Jupiter producing an anticyclonic storm. It is thrice the size of the planet Earth. Similarly, the planet Neptune has the Great Dark Spot.



The Neptune is a giant planet and eighth planet in the solar system. It is more than 30 times as far from the Sun as our Earth. It is the most distant planet orbiting the Sun and is not visible to the naked eye.

The Great Dark Spot on the planet Neptune is a huge storm in its southern hemisphere atmosphere. The size of this spot is about the size of our Earth. The wind speed of this storm is about 1,500 miles per hour. The spot looks dark because the air particles darkening in a layer below the main visible haze atmospheric layer as ice and haze mix together in Neptune's atmosphere.

The Great Dark Spot was first discovered when Voyager-2 spacecraft flew by Neptune in 1989. But when the Hubble Telescope looked at Neptune in 19994, the Great Dark Spot was not there as before and a different dark spot was visible in the northern hemisphere atmosphere of the Neptune. Unlike Jupiter's great red spot, which has lasted for hundreds of year, the life time of the great dark spots appear to be shorter, forming and disappearing once in every few years.

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AMATEUR ASTRONOMY

Rabindranath Swain
Science Communicator Puri

Amateur astronomers can build their own equipment, and hold star parties and gatherings, such as Stellafane .

Astronomy is one of the sciences to which amateurs can contribute the most. Collectively, amateur astronomers observe a variety of celestial objects and phenomena sometimes with consumer-level equipment or equipment that they build themselves. Common targets of amateur astronomers include the Sun, the Moon, planets, stars, comets, meteor showers, and a variety of deep-sky objects such as star clusters, galaxies, and nebulae. Astronomy clubs are located throughout the world and many have programs to help their members set up and complete observational programs including those to observe all the objects in the Messier (110 objects) or Herschel 400 catalogues of points of interest in the night sky. One branch of amateur astronomy, astrophotography, involves the taking of photos of the night sky. Many amateurs like to specialize in the observation of particular objects, types of objects, or types of events that interest them.

Most amateurs work at visible wavelengths, but many experiment with wavelengths outside the visible spectrum. This includes the use of infrared filters on conventional telescopes, and also the use of radio telescopes. The pioneer of amateur radio astronomy was Karl Jansky, who started observing the sky at radio wavelengths in the 1930s. A number of amateur astronomers use either homemade telescopes or use radio telescopes which were originally built for astronomy research but which are now available to amateurs (*e.g.* the One-Mile Telescope). Amateurs can make occultation measurements that are used to refine the orbits of minor planets. They can also discover comets, and perform regular observations of variable stars. Improvements in digital technology have allowed amateurs to make impressive advances in the field of astrophotography.

AMETURE ASTRONOMY IN INDIA

The Confederation of Indian Amateur Astronomers (CIAA) is a national level organisation of amateur astronomers in India that convenes a national meeting of members every year, and coordinates the activities of amateur astronomers throughout the country. One can join an amateur astronomers club in India by searching online for clubs in the local area. Some popular clubs in India include the Indian Astronomical Society, the Bangalore Astronomical Society, Delhi Astronomy Club & Samanta Chandrasekhar Ameture Astronomers Association (SCAAA) Bhubaneswar.

The Astronomical Society of India was established in 1972 and has grown to become the prime association of professional astronomers in India. The society has close to 1000 members. The objectives of the society are the promotion of Astronomy and related branches of science in India.

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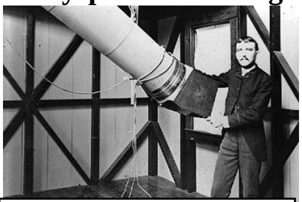
Imaging in Astronomy – a short history.

Dr. Nihar Pattnaik, nihardwellshere@gmail.com

What does a piece of rock have in common with a 200+ megabyte RAW image downloaded from the website of the Webb telescope ? We try and find the answer to this question in the following few paragraphs. The history of astrophotography unfolds like a cosmic tapestry, intricately woven with threads of human curiosity, technological innovation, and a relentless quest to capture the elusive beauty of the heavens. Spanning centuries, this journey reflects not only the evolution of photographic techniques but also the profound impact on our understanding and appreciation of the cosmos.

Humble Beginnings - From the earliest stone carvings found in rocks all over the world to the latest images from the James Webb Space Telescope, humanity has always wanted a pictorial representation of the night sky. Cave paintings show up all over the world from the paleolithic and nneolithic perios, often representing constellations that are still easily recognisable. A pictorial representaion of the night sjy is arguably the most efficient way of record keeping. This tradition continued past the medieval, renaissance periods and well into the middle of the Industrial revolution where a new kind of revolution was being born in the darkness of a photographic developing room.

Early pioneers – Daguerreotypes and wet plates : The nascent stages of astrophotography emerged in the 19th century, a period when astronomy and photography were in their infancy. The invention of the daguerreotype in 1839, a process involving light-sensitive silver-coated copper plates, marked a pivotal moment in the history of photography. Inspired by this breakthrough, astronomers sought to harness this new technology to immortalize their observations. In the 1840s, scientists like John William Draper and Henry Draper began experimenting with daguerreotypes to capture the moon and stars. These early attempts, while rudimentary by today's standards, laid the foundation for a revolutionary fusion of art and science. Capturing long-exposure images on these plates allowed astronomers to record celestial details with a permanence previously unattainable.

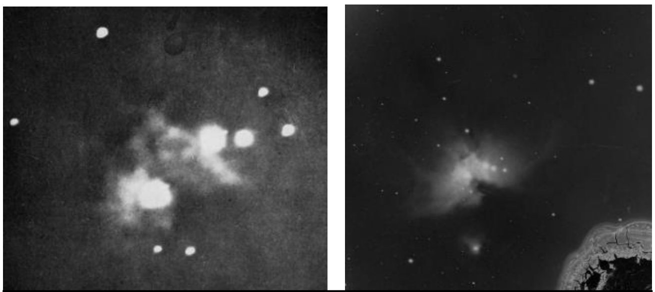


Henry Draper

Astrophotography then met its first major hurdle. Film and plates were not sensitive enough to capture starlight in a momentary exposure. They needed to be exposed for longer durations. The problem is, the earth rotates, making the entire sky appear to change with time. Any exposure left open to the stars will only show streaks over the time it takes the stars to cross its field of view. This necessitated the need for accurate tracking.

The first tracking mounts were essentially mechanical clocks powered by a wound up spring, with an axis perfectly aligned with the earths own axis of rotation. Any telescope placed on such a mount, would automatically track the stars as they move across the sky. The speed of this telescope would be precisely regulated by a device called a speed governor, and many such telescopes can still be seen today in perfectly preserved condition.

As the 19th century progressed, wet plate photography supplanted daguerreotypes in the field of astrophotography. The wet plate process, utilizing glass plates coated with a light-sensitive emulsion, provided a more flexible and reproducible medium. Pioneering astronomers like Edward Emerson Barnard, with his keen eye and meticulous technique, made significant contributions during this era. Barnard's photographic plates revealed the intricate structure of the Milky Way and unveiled dark nebulae, previously concealed by the limitations of visual observation. Around this time, telescope mounts were being powered by DC electricity , making tracking more precise.



First picture of the Orion nebula (left) taken in 1882&a considerable better photograph (right) following year.

20th Century advancements – Photomultipliers and CCDs

The turn of the 20th century witnessed a quantum leap in astrophotography with the introduction of electronic imaging devices. Photomultiplier tubes, capable of detecting faint light, became integral to astronomical research. Their sensitivity allowed astronomers to explore the depths of the cosmos with unprecedented precision.

A pivotal moment came with the development of charge-coupled devices (CCDs) in the 1960s. These semiconductor-based devices replaced traditional photographic plates and offered advantages such as higher sensitivity, efficiency, and the ability to store and digitally process images. The shift to CCD technology marked a transformative era, as astronomers could now capture fainter celestial objects and conduct more intricate studies of the universe.

Hubble and Beyond

Perhaps the most iconic symbol of modern astrophotography is the Hubble Space Telescope, launched into orbit in 1990. Equipped with a suite of scientific instruments, including high-resolution cameras, Hubble revolutionized our understanding of the cosmos. Its ability to capture stunning images of distant galaxies, nebulae, and other celestial wonders has not only expanded our scientific knowledge but also captivated the imaginations of people worldwide.

The Hubble Space Telescope's breathtaking images have become cultural touchstones, appearing in textbooks, documentaries, and even popular media. Its contributions extend beyond scientific discovery, serving as a bridge between the scientific community and the general public, fostering a sense of wonder and curiosity about the universe. The James Webb Telescope with its iconic six-pointed stars is the latest to join this ever growing era of large , sensitive space telescopes that usher our species into an exciting time of discovery. From the farthest galaxies to exoplanet atmospheres , the Webb has proven to be a workhorse instrument in every niche field of modern astronomy.

Photograph of the central region of Orion nebula, Hubble/Webb composite

The Digital revolution :

In recent decades, the digital revolution has democratized astrophotography, making it accessible to amateur astronomers and enthusiasts. The advent of affordable digital cameras and sophisticated imaging equipment has transformed backyard observatories into hubs of celestial exploration. Amateurs armed with telescopes, specialized cameras, and image processing software can now capture awe-inspiring views of the night sky, rivaling the images produced by professional observatories just a few decades ago. It is now very much within the realm of possibility to have a rooftop or backyard setup, armed with a simple star tracker and a camera lens and produce portrait quality images of galaxies millions of light years away. The hobby of astrophotography is both an art and a science, with a very rewarding learning curve, and a complete novice can start producing beautiful images with just a few nights of practice.

Online communities and social media platforms have played a pivotal role in this democratization process. Platforms like Instagram, Twitter, and dedicated astrophotography forums like Astrobin and Cloudynights serve as virtual spaces where enthusiasts share their images, exchange tips and techniques, and inspire others to join the cosmic pursuit. The global network of amateur astrophotographers has become a vibrant community, fostering collaboration and camaraderie across borders.

Image of the orion nebula taken by the author, 4 inch refractor and DSLR camera

Challenges and Future Frontiers

Despite the remarkable progress in astrophotography, challenges persist. Light pollution, the interference of artificial light from human activities, poses a significant obstacle for both amateur and professional astronomers. Efforts to combat light pollution range from advocacy for responsible outdoor lighting practices to the development of advanced filters and technologies designed to mitigate its impact.

As technology continues to advance, new frontiers in astrophotography beckon. Emerging technologies such as adaptive optics, which correct distortions caused by Earth's atmosphere in real-time, promise sharper and clearer images. A recent trend in this field is the rapid availability of the so called “Smart telescopes”. Though not as versatile as a dedicated setup where one can change out parts and components to their liking, a smart telescope offers the option to “goto”any object in the night sky and immediately view the target wirelessly on a smartphone. These telescopes though still somewhat expensive and still in their infancy, offer the convenience of not having to setup, being extremely portable, and offer near-instant imaging capabilities. They are the perfect choice for outreach & planetarium clubs for doing “quick: and low quality astrophotography from within the city.

Through the lens into the future

The history of astrophotography is a testament to the human spirit of exploration and the relentless pursuit of knowledge. From the early daguerreotypes capturing the moon's craters to the digital images of distant galaxies captured by today's amateurs, each photograph is a snapshot of our cosmic odyssey.

Astrophotography has not only deepened our understanding of the universe but has also ignited a universal sense of wonder and awe. The images captured by astronomers, whether professional or amateur, serve as windows to the cosmos, inviting us to contemplate the vastness of space and our place within it.

As we stand on the shoulders of those early pioneers who pointed their lenses towards the stars, we recognize that the history of astrophotography is an ongoing narrative. With each new discovery, technological breakthrough, and shared image, we continue to unravel the cosmic mysteries that have fascinated humanity for centuries, embarking on a journey that extends beyond our terrestrial boundaries and into the limitless expanse of the cosmos.

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Navigating the Cosmos through Data The Adventure of Data-Driven Astronomy

Chiranjibi Kunda, Member, SCAAA

Astronomy has always been akin to an explorer setting sail into an unknown ocean, with stars as distant lighthouses and galaxies as uncharted islands in the cosmic sea. But in today's world, our vessels in this cosmic sea are not just telescopes but also by the incredible might of computers and the wealth of data they process. We're now in the age of data-driven astronomy, a time where data acts like a compass, guiding us to new and exciting celestial discoveries.

Imagine the universe as a colossal library. Each book represents a celestial object, filled with stories of its past, present, and potential future. For centuries, astronomers were like old-school librarians, meticulously cataloging these books one at a time. But now, imagine a super-efficient librarian, equipped with the latest technology, who can scan and analyze thousands of books in mere seconds. This is what data-driven astronomy does; it harnesses the power of advanced computing to sift through the immense volumes of data gathered by telescopes.

This shift in astronomy really picked up steam with the introduction of digital imaging. We moved from traditional photographic plates to using CCDs (Charge-Coupled Devices) in telescopes, marking the dawn of a new, digital era in stargazing. These CCDs work like the sensor in a digital camera, capturing light and converting it into electronic data that can be stored, analyzed, and interpreted by computers.

To grasp the sheer magnitude of data (often referred to as 'big data') produced by astronomical telescopes, consider the Large Synoptic Survey Telescope (LSST). This marvel of modern astronomy is anticipated to generate an astounding 3 terabytes of data every single night. To put this into perspective, that amounts to a colossal one petabyte (equivalent to 1,000 terabytes) every two days. Remarkably, this is akin to the total volume of internet traffic in Canada over the same timeframe! Such figures not only highlight the technological prowess of the LSST but also underscore the exponential growth of data in the field of astronomy.

Modern telescopes, such as the Hubble and recently launched James Webb Space Telescope, are like ultra-high-resolution cameras. They capture detailed images of the universe, with every pixel potentially holding secrets about distant planets and ancient galaxies. But with millions of galaxies out there, manually checking each image just isn't feasible. This is where data-driven astronomy steps in, using sophisticated algorithms and machine learning techniques to quickly analyze these images, spotting patterns or anomalies that might point us to something new or unexpected.

A prime example of this is in the search for exoplanets. Before, finding planets outside our solar system was incredibly challenging, akin to trying to spot a firefly next to a glaring spotlight. Earlier techniques involved looking for tiny changes in a star's position or brightness due to the influence of a planet. Now, thanks to data-driven methods, we can detect these distant worlds by looking at minuscule variations in starlight, which are captured in the massive datasets from telescopes. Sophisticated algorithms parse through this data, picking up on the faint signals of exoplanets, something almost impossible for the human eye to do. It's like we've developed a new sense, enabling us to hear the faintest whispers of these far-off planets. But this isn't a journey we're on alone. Data-driven astronomy is a collaborative adventure, involving astronomers, Data scientists and Data engineers, and even citizen scientists. Through online platforms like Zooniverse, RAD@Home India, etc people from all walks of life can dive into astronomical data, helping to classify galaxies or identify new celestial events. These joint efforts not only speed up scientific discovery but also make it more inclusive, allowing anyone with an interest and an internet connection to be a part of groundbreaking research.

In summary, data-driven astronomy is revolutionizing how we understand the universe. It's a fusion of ancient cosmic light and modern technology. As we continue to sail this star-filled ocean, armed with our data compass, the mysteries of the universe unfold before us. With each new dataset, algorithm, and collaborative effort, we inch closer to unraveling the enigmas of the cosmos. The universe is speaking to us, and now, more than ever, we have the tools to listen and understand.

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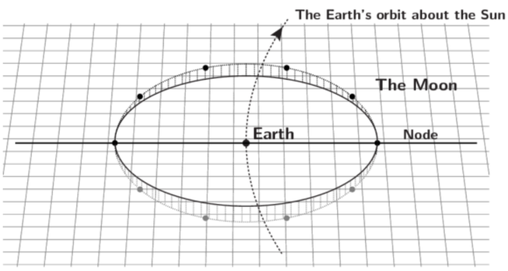
MYTH AND MATH BEHIND RAHU-KETU

Ar. Jagadish Prasad Tripathy



In Hindu mythology and astronomy, the celestial entities Rahu and Ketu hold a unique and intriguing place. These mythical figures are considered to be the head and tail of a snake and play a significant role in shaping the cosmic narrative. Understanding their mythological and astronomical significance provides a deeper insight into the rich tapestry of Hindu beliefs and the cosmic dance of celestial bodies.

Mythological Significance: According to Hindu mythology, Rahu and Ketu are shadowy planets that play a pivotal role in the cosmic drama of churning the ocean of milk, known as Samudra Manthan. The story goes that the Devas (celestial beings) and Asuras (demons) joined forces to churn the ocean, seeking the elixir of immortality known as Amrita. As the elixir emerged, Lord Vishnu assumed the form of Mohini, a bewitching enchantress, to distribute it among the Devas.

However, Swarbhanu, an Asura in disguise, managed to sneak between the sun and the moon into the line of Devas and drink a drop of the Amrita. Sun and Moon, who witnessed this deceit, alerted Lord Vishnu, who promptly severed Swarbhanu's head with his Sudarshana Chakra. As a result, Swarbhanu became immortal, but his body was split into two parts – the head, known as Rahu, and the tail, known as Ketu. Logical questions may arise at this point. Why did Swarbhanu sneak between the sun and moon? Why are Rahu and Ketu polar opposites in the sky, and why is the body of a snake attached to Rahu and Ketu? The answer lies in astronomy.

Astronomical Significance: The orbital plane of the moon around the Earth is inclined to the orbital plane of the Earth around the sun (ecliptic plane) by 5° 9'. The intersection between both planes forms a finite line with two endpoints (nodes) which divides the lunar orbit into two equal halves. The node after crossing which the moon appears to move slightly northwards from the ecliptic is known as the North node or Rahu, and the other point is known as the South node or Ketu. The nodal axis, connecting these two points, plays a crucial role in the occurrence of eclipses. When the Sun, Moon, and Earth align with the nodal axis on a full moon or new moon day, the celestial drama unfolds, creating the visually striking phenomenon.

The mathematical precision of Rahu and Ketu in predicting eclipses is a testament to the precision of celestial mechanics. A period of approximately 18.6 years is the time in which these nodes complete one full transit over the ecliptic or Rashi chakra by staying 18 months in each sign. This cycle results from the alignment of the Sun, Earth, and Moon in nearly the same relative geometry, causing a repetition of similar eclipse patterns.

As we unravel the astronomical and mathematical significance of Rahu and Ketu, we gain a profound appreciation for the intricate knowledge possessed by ancient civilizations. These celestial nodes continue to bridge the gap between the cosmic and the terrestrial, inviting further exploration and study into the fascinating realms of astronomy and mathematics intertwined with the rich cultural heritage of Hinduism.

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Amateur Astronomy – my personal journey

Kishore Chandra Pattanaik, Member, SCAAA

My journey commenced with an insatiable curiosity about the mysteries of existence and a longing for a connection with something greater than myself. I delved into religious texts and philosophical teachings, I found that many spiritual traditions were intertwined with cosmic symbolism and references to celestial bodies.

Intrigued by these celestial connections, I decided to immerse myself in the study of astronomy. The night sky, adorned with stars, planets, and galaxies, became my sacred text. It was under this cosmic canopy that I sought to unravel the secrets of the universe, hoping to catch a glimpse of the divine order that might lie hidden within the cosmic ballet.

In 2016, I decided to immerse myself in the quest of amateur astronomy. I managed to finally find the time I had long needed and joined SCAAA as a club member, coming into contact with various other members . Around this time, I also purchased my very first telescope- a 10 inch dobsonian that I cherish to this day.

Staring through the eyepiece of the telescope, I marveled at the intricate patterns of constellations, each telling a story that transcended earthly boundaries. The vastness of space, with its infinite galaxies and the dance of celestial bodies, invoked a sense of awe and humility, prompting me to contemplate the grandeur of creation.

As I started following more resources online and delved deeper into the science of astronomy, I discovered the hobby of Astrophotography – a logical extension of visual astronomy taken up a notch higher. A DSLR at first, a dedicated astro camera, a new refractor, a tracking mount etc soon followed. A full fledged observatory now rests on my rooftop and it gives me much joy to entertain curious visitors.

The telescope, once a tool for scientific exploration, became a conduit for my spiritual journey. Through its lens, I glimpsed the majesty of the universe and pondered the intricate tapestry of creation. The vastness of the cosmos became a canvas, and each celestial body, a stroke of divine artistry.

In the quiet moments spent gazing at the night sky, I felt a profound connection with something beyond the tangible world. The stars, far from being mere luminous dots, became emissaries of the divine, guiding me on my quest for a deeper understanding of God.

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Measuring height of Kapilash from Cuttack

Dr. Subhendu Pattnaik,
Ex-Deputy Director, Pathani Samanta Planetarium

We are well aware of Samanta Chandra Sekhar, popularly known as Pathani Samanta in Orissa, for his contribution in astronomy following traditional methods. He was not aware of the telescope and other instruments developed in the west. He took observations with ingenious and handy instruments, all fabricated by himself. His study and observations are recorded in an invaluable classic, the Siddhanta Darpana. The most talked episode about Samanta's contribution was instant calculation of angular separation of two stars and correct prediction of height of Kapilash Mountain from Cuttack by using his handmade instruments. It is mentioned number of times that the height of Kapilash was calculated by using Manayantra a 'T' shaped instrument. Detail explanation are available in various platforms regarding use of the Manayantra and Trigonometry behind the process. We in past have used those instruments and methods to calculate the height of nearby objects like buildings and verified the results. Which came out to be within permissible error limits. However, while recently working on to find out the truth regarding a photograph of Puri Jagganath temple visible from Bhubaneswar due to reduction in pollution during covid. The photograph is being circulated in social media widely for many days. However, I got some shocking results while trying to find the probability of visibility of Puri Temple from Bhubaneswar.

In the social media the photo in question shows a nice picture of Puri Jagganath temple taken from Master canteen Bhubaneswar. Which was followed by lot of comments regarding whether the photo is real of fake. Although there was no scientific backing of respective claims, I tried to find out the truth and tried to prove / disprove the matter scientifically from various angles. The question was "can we see Puri Jagganath temple from Bhubaneswar with naked eye?"

While making all these calculations one thing struck to my mind which we had heard/read number of times regarding Patha Samanta, calculating height of Kapilash mountain from Cuttack correctly. Was the mountain actually visible from Cuttack or not and what would have been its apparent size of the mountain and how he calculated the correct height worried me. After solving the Puri temple problem, I tried to verify the facts in same line. We presume that the atmosphere is absolutely clear and there is no hindrances ie tall building/ trees in between. At the initial stage let us assume Earth is flat enough and there is no curvature effect for a distance of 40 KM between Cuttack and Kapilash.

Case study – 1

Assumptions :

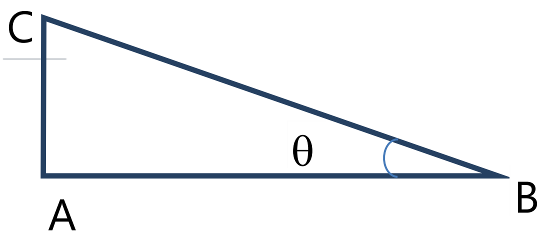
a. atmosphere is absolutely clear with highest visibility
b. no hindrance ie tall building/ trees in between.
c. Curvature of Earth is zero ie Cuttack and Kapilash are in same level and line of site is parallel to ground.

AB = Distance Between Cuttack and Kapilash = 40 KM = 40,000m

AC = Height of Kapilash Mountain = 2150 ft = 655 m

Angle ABC = θ

Hence, Tan θ = AC/AB



For long distance objects, using the following information,
1 degree = 60 min = 3600 arc sec; and
degree = 57 radian.
Angular Diameter (θ) in degrees = 57 × (Actual size ÷ distance)

Angular size in degree = 57(655 ÷ 40,000)
= 0.933375 degree ≈ 60 arc sec ie double the apparent size of full moon.

In night sky the full moon which is at a distance of 3.84 lac Km from us seen with an angular diameter of 30 arc min. Imagine the apparent size of the full moon which will be around 1 foot in diameter. So, the apparent size of moon with 1 foot in diameter sustains an angular diameter of 30 arc min in our eye. That means the apparent height of Kapilash mountain to naked eye will be around 2 ft only.

Now I have a serious confusion/concern that how Pathani Samanta estimated height of Kapilash mountain from Cuttack with such a small apparent size of 2 ft only. That has to be explored and answered scientifically and may be doing practically.

Case study – 2

Here I would like to give an idea of resolving power of human eye. That means how small can our eye see in angular diameter.

The angular resolution, which is also called spatial resolution of any optical system can be estimated by Rayleigh's Criterion. When two point sources are resolved from each other, they are separated by at least the radius of the airy disk.

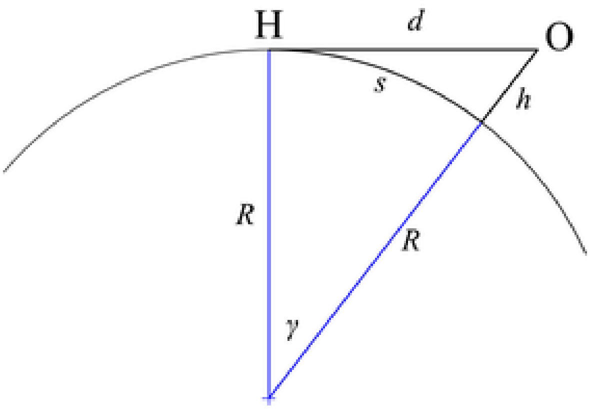
When $\Theta = 1.22\,(\lambda/D)$ rad ,

where Θ is the angular resolution, λ is the wavelength of light and D is the diameter of the eye.

We know that that 360 degrees = 2π radians.

The eye pupil diameter changes during day and night, whereas the day the pupil size is between 3 mm to 4 mm and at night it is from 5 mm to 9 mm. So according to Rayleigh's Criterion, we can calculate the spatial resolution of human eye.

Lets say that at day time the pupil size is 3 mm and the optimal sensitivity is 0.55 µm, (wavelength of light) we can apply the rule.



$\Theta = 1.22(\lambda/D)$rad

$=1.22(0.55\ \mu m/3\ mm)$rad$(180\ deg/\pi\ rad)(1\ mm/10^3\ \mu m)$

= 0.0128 deg(3600”/1 deg)

= 50” (arc second) (day time)

Moreover, at night the pupil diameter increases to 9 mm to increase the observation, we can do the same calculation to find the angular resolution of the eye at night.

So in day time human eye can resolve or see objects as small as 50 arc second.

Case study – 3

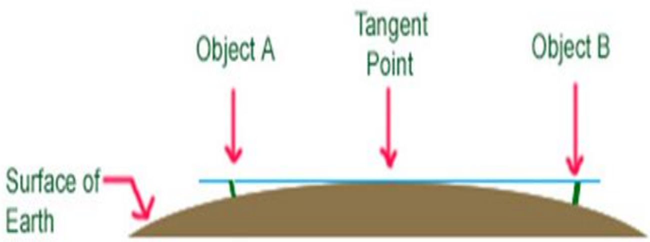
Till now we have not considered the curvature of earth surface. We have assumed it to be flat in our previous cases, which is not correct. When one of my friends who is in Indian navy suggested that the curvature of earth surface also plays a major role in this case, I started exploring the idea. You will be surprised to know that even though the distances are very small, curvature of earth surface plays a vital role while answering our original question.

From the Figure by using simple trigonometry we can say that

$2Rh = d^2$ Or $h = \frac{d^2}{2R}$

We now 2R = Diameter of earth = 12742 Km

Hence, h (in m) = 0.0785 d^2(in km)



Taking distance of Kapilash from Cuttack as 40 Km we can find that

$h_{(kapilash)}$ = 125.6m ≈ 412 ft

Hence Kapilash is 412 ft below the horizon line of sight from Cuttack. That means the curvature of earth surface is also playing a major role in these cases.

Here we have to recalculate the 2nd case as Kapilash is below 412ft line of sight from Cuttack. That means the height of Kapilas taken as 2150 ft is to be reduced by 412 ft. Hence in new calculation it can be found that the apparent height of Kapilash from Cuttack is

Height of Kapilash Mountain = 1738 ft = 530 m,

Areal distance from Cuttack = 40 km = 40,000m

Angular size in degree = 57(530 ÷40,000) = 0.755 degree
≈ 45 arc minutes

Equivalent to apparent height of 1ft 6 inches.

This has given rise to a serious question and re thinking of the fact that, how and using what method (if any ?) Pathani Samanta had calculated the height of Kapilash mountain from Cuttack using two sticks only. Please explore the answer, redo the calculations and let me know or give me some time to find out the answer by consulting some experts.

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Samanta Chandrasekhar Jyotirbigyani Sammana- 2023-24

Prof. Lambodar Prasad Singh, Retired Professor in Physics, Utkal University

Sir,

You are an eminent physicist associated with teaching for more than 4 decades. You had the opportunity to conduct research in renowned institutes such as Germany's Dortmund and Ulm Universities, Italy's Abdus Salam International Center for Theoretical Physics, Geneva-based European Nuclear Research Center and Ireland's Dublin Institute of Advanced Studies. Your contribution in the field of scientific research and creation of popular literature is unique. . You had held many responsibilities such as Dean of the Faculty of Science, Head of the Department of Physics at the University of Utah. Your books written in Odia and translations of books by Nobel laureate Erwin Schrödinger and renowned mathematician G. H. Hardy are widely read. You have interests in popularizing science and received awards and accolades from various prestigious institutions.

Samanta Chandrasekhar Amateur Astronomers Association (SCAAA) Bhubaneswar feels proud to confer you the Samanta Chandrasekhar Jyotirbigyani Sanmana for the year 2023-24 and wishing you a long, healthy and bright future in life.

Samanta Chandrasekhar Jyotirbigyani Sammana- 2023-24

Prof. Ajit Mohan Srivastav, Retired Professor in Physics, Institute if Physics

Sir,

You are an eminent physicist associated with research work for more than 4 decades. You have completed B.Sc. 1981 from Allahabad University, M.Sc. 1983 in Physics Indian Institute of Technology, Kanpur & Ph.D. 1989, in Theoretical High Energy Physics, Syracuse University Syracuse, New York, USA.

You have contributed immensely as Research Associate at Theoretical Physics Institute University of Minnesota, Minneapolis, Minnesota, USA from 1989 – 92, Research Associate, Institute for Theoretical Physics, University of California Santa Barbara, California, USA from 1992 – 94 and Faculty at Institute of Physics, Bhubaneswar, 1994 – 2022.Your Area of research are Elementary particle physics, cosmology, quark-gluon plasma, pulsars, gravitational waves, topological defects, liquid crystal experiments

Your contribution in the field of scientific research and popularisation of Science is unique. You have received awards and accolades from various prestigious institutions.

Samanta Chandrasekhar Amateur Astronomers Association (SCAAA) Bhubaneswar feels proud to confer you the Samanta Chandrasekhar Jyotirbigyani Sanmana for the year 2023-24 and wishing you a long, healthy and bright future in life.

Samanta Chandrasekhar Jyotirbigyani Sammana- 2023-24

Dr. Subhendu Pattnaik,Ex-Dy. Director, Pathani Samanta Planetarium.

Sir,

You are a true scientist, science writer and excellent science communicator. Born in 1963 you completed 10th in 1979 and +2 (Science) in 1981, from DM School, obtained degree in Electrical Engineering from CET and joined Pathani Samantha Planetarium in the year 1988. Your interest and passion towards astronomy led you to pursue PhD related to Astronomy and awarded with PhD degree in Physics during the year 2013.

Your contribution in popularizing science and attracting students towards science stream is unique. Your efforts to remove the superstitions associated with Eclipse and other celestial events are commendable. Your publications of articles and books are highly appreciated by one and all. You have received many awards and accolades from various prestigious institutions.

Samanta Chandrasekhar Amateur Astronomers Association (SCAAA) Bhubaneswar feels proud to confer you the Samanta Chandrasekhar Jyotirbigyani Sanmana for the year 2023-24 and wishing you a long, healthy and bright future in life.

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